# Week 6 Lab Activity

## Week 6 Lab Activity: Multiple Factors

Upload your data to the class google sheet which will be made available! Once it's ready, be sure to download it as a CSV or otherwise make sure it's available to you later.

In this lab, we want to investigate how forest edge conditions shape herb-layer plant richness.

This leads us to our study question: How do canopy openness, light intensity, litter depth, tree presence, and edge distance influence herb-layer plant species richness?

The dataframe we generated with our data contains several different variables:

- ID = sample number (label, not used in our analysis)
- transect = A, B, C, D (label, not used in our analysis)
- edge_m = distance to edge (m)
- richness = species count (number of species)
- lux = light intensity/PAR (transform later) (PPDF or umol/m2/s)
- log10_lux = light intensity, log base 10 transformation
- cover = canopy cover (percent)
- tree_m = nearest tree (m)
- litter_cm = litter depth (cm)
- notes = comments (not used in our analysis)

# In JASP:

1. Import the data, adjust column types as needed.
2. Descriptives → Distribution Plots for all variables. (1 point)
    a. Comment on what variables you might transform using a log transformation, if any. No need to do the transformation now. (1 point)
3. Regression → Correlation → Create a correlation matrix (Pearson) by adding all variables. Provide scatter plots and a table. I find that unchecking "Display pairwise" aids legibility. (1 point)
    b. Do you think there are any variables that are particularly strongly correlated with species richness? If so, which one(s)? (1 point)
4. Regression → Linear Regression → Create a model with richness as your dependent variable and log10lux, cover, tree_m, and

litter_cm as covariates. Display the model fit, regression equation, coefficient estimates, and tolerance and VIF. Additionally, display the residuals vs predicted plot, and the QQ plot. (2 points)

    c. Are there any variables in this model that are statistically significant? If you had to pick one variable to remove from the model, which one would you choose and why? (1 point)
    d. Is there any evidence of multicollinearity, and how can you tell? (1 point)

5. Regression → Linear Regression → Create another model, this time adding in edge_m as a covariate. Display the model fit, regression equation, coefficient estimates, and tolerance and VIF. Additionally, display the residuals vs predicted plot, and the QQ plot. (1 point)

    e. Compare the R2 and adjusted R2 values between this model and the

last one. Which one might be a better model to use? (1 point)

f. In looking at your model coefficients, you can see that there are unstandardized and standardized coefficients. The unstandardized ones are those that you can see in your regression equation, and you're able to use them to predict a value for your dependent variable given measurements of all your other variables. Standardized coefficients have no units, and give you information on how many standard deviations a dependent variable will change per standard deviation increase in the covariate. Knowing this, which predictor has the strongest standardized effect? (1 point)

6. When looking at your diagnostics and your latest model, if you would have performed

a transformation earlier or maybe removed a covariate in your model, try that now and compare it to the model from part 5. If you would not have done these things, reiterate why. (2 points)

7. Lastly, what does a coefficient mean in ecological terms (holding others constant)? You can use an example from your model to illustrate. (1 point)

# Deliverables

PDF (or otherwise) answering the questions and displaying your JASP results.

This will be due on Thursday, 10/08, at 11:59 PM.